

PRACTICAL SESSION-2

TASK :

```
((base) saiabburi@Geyasree-MacBook-Air ~ % mkdir Session2
mkdir: Session2: File exists
((base) saiabburi@Geyasree-MacBook-Air ~ % cd Session2
((base) saiabburi@Geyasree-MacBook-Air Session2 % nano src/file_copy.c
((base) saiabburi@Geyasree-MacBook-Air Session2 % mkdir -p src
((base) saiabburi@Geyasree-MacBook-Air Session2 % nano src/file_copy.c
((base) saiabburi@Geyasree-MacBook-Air Session2 % gcc -Wall -Wextra -o file_copy src/file_copy.c
((base) saiabburi@Geyasree-MacBook-Air Session2 % echo "This is a sample file for the OSSP file copy task." > sample.txt
((base) saiabburi@Geyasree-MacBook-Air Session2 % cat sample.txt
This is a sample file for the OSSP file copy task.
((base) saiabburi@Geyasree-MacBook-Air Session2 % This is a sample file for the OSSP file copy task.
zsh: command not found: This
((base) saiabburi@Geyasree-MacBook-Air Session2 % ./file_copy sample.txt copied.txt
File copied successfully.
Source : sample.txt
Destination: copied.txt
((base) saiabburi@Geyasree-MacBook-Air Session2 % cat copied.txt
This is a sample file for the OSSP file copy task.
((base) saiabburi@Geyasree-MacBook-Air Session2 % cp sample.txt copied.txt
((base) saiabburi@Geyasree-MacBook-Air Session2 % cat copied.txt
This is a sample file for the OSSP file copy task.
```

UW PICO 5.09

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s <source_file> <destination_file>\n", argv[0]);
        return 1;
    }

    int source_fd = open(argv[1], O_RDONLY);

    if (source_fd < 0) {
        perror("Error opening source file");
        return 1;
    }

    int destination_fd = open(
        argv[2],
        O_WRONLY | O_CREAT | O_TRUNC,
        0644
    );

    if (destination_fd < 0) {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    char buffer[BUFFER_SIZE];
    ssize_t bytes_read;

    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0) {
        ssize_t bytes_written = write(
            destination_fd,
            buffer,
            bytes_read
        );

        if (bytes_written != bytes_read) {
            perror("Error writing to destination file");
            close(source_fd);
            close(destination_fd);
            return 1;
        }
    }
}
```

```
char buffer[BUFFER_SIZE];
ssize_t bytes_read;

while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0) {

    ssize_t bytes_written = write(
        destination_fd,
        buffer,
        bytes_read
    );

    if (bytes_written != bytes_read) {
        perror("Error writing to destination file");
        close(source_fd);
        close(destination_fd);
        return 1;
    }
}

if (bytes_read < 0) {
    perror("Error reading source file");
}

close(source_fd);
close(destination_fd);

printf("File copied successfully.\n");
printf("Source : %s\n", argv[1]);
printf("Destination: %s\n", argv[2]);

return 0;
```

```
}
```

PRACTICE TASKS:

TASK 1:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
        printf("Executing command: %s\n\n", command);

        execlp(command, command, NULL);

        printf("Command execution failed.\n");
    }
    else if (pid > 0)
    {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }
    else
    {
        printf("Fork failed!\n");
    }

    return 0;
}
```

```

(base) saiabburi@Geyasree-MacBook-Air OSSP % gcc command_executor.c -o command_executor
(base) saiabburi@Geyasree-MacBook-Air OSSP % ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 7940
Child PID : 7941

Child Process
Child PID : 7941
Parent PID : 7940
Executing command: ls

LICENSE          assets          command_executor  fork_exec        obj              tests
Makefile         bin            command_executor.c  fork_exec.c      scripts
README.md        command_executor.c  docs              include          src

```

Child process completed.

```

(base) saiabburi@Geyasree-MacBook-Air OSSP % █

```

TASK 2:

Uname -a :

```

Last login: Tue Aug 11 22:20:02 on ttys000
(base) saiabburi@Geyasree-MacBook-Air ~ % cd OSSP
(base) saiabburi@Geyasree-MacBook-Air OSSP % uname -a
Darwin Geyasree-MacBook-Air.local 22.5.0 Darwin Kernel Version 22.5.0: Thu Jun  8 22:21:34 PDT 2023; root:xnu-8796.121.3~7/RELEASE_ARM64_T8112 arm64

```

Lscpu :

Ps:

```

Darwin Geyasree-MacBook-Air.local 22.5.0 Darwin Kernel Version 22.
(base) saiabburi@Geyasree-MacBook-Air OSSP % lscpu
[zsh: command not found: lscpu]
(base) saiabburi@Geyasree-MacBook-Air OSSP % ps
[
  PID TTY          TIME CMD
  8285 ttys000      0:00.03 -zsh

```

Ps aux :

(base) saiabburi@Geyasree-MacBook-Air OSSP % ps aux										
USER	PID	%CPU	%MEM	VSZ	RSS	TT	STAT	STARTED	TIME	COMMAND
saiabburi	7621	7.1	4.7	1922182880	393504	??	S	10:08PM	2:28.12	/Applications/Go
_windowserver	372	1.6	0.8	410722144	63216	??	Ss	8:28AM	77:40.12	/System/Library/
saiabburi	1445	0.9	2.4	1943926944	201632	??	S	8:36AM	6:39.68	/Applications/Go
saiabburi	921	0.8	1.1	460556720	94928	??	S	8:28AM	27:27.64	/Applications/Go
saiabburi	6568	0.8	4.7	1947840288	395248	??	S	7:52PM	6:01.91	/Applications/Go
saiabburi	904	0.5	3.7	510055488	313152	??	S	8:28AM	26:44.46	/Applications/Go
saiabburi	8299	0.5	1.6	1923810048	130176	??	S	10:35PM	0:00.25	/Applications/Go
saiabburi	7873	0.4	0.6	409887456	51744	??	S	10:22PM	0:06.32	/System/Applicat
saiabburi	7250	0.3	0.2	409241552	18096	??	S	9:55PM	0:02.05	/System/Library/
saiabburi	7645	0.2	0.2	408284256	17152	??	S	10:08PM	0:02.90	/System/Library/
root	342	0.1	0.1	408269024	9872	??	Ss	8:28AM	0:25.63	/System/Library/
root	366	0.1	0.1	408264528	9104	??	Ss	8:28AM	1:46.56	/usr/libexec/cor
saiabburi	925	0.1	1.0	459102208	85376	??	S	8:28AM	17:19.24	/Applications/Go
saiabburi	7986	0.1	1.0	414513312	81696	??	S	10:28PM	0:09.11	/Applications/Gr
saiabburi	6289	0.1	0.1	408103072	4368	??	S	7:11PM	0:01.91	/System/Library/
saiabburi	7921	0.1	0.9	1923835712	74352	??	S	10:24PM	0:01.23	/Applications/Go
_mysql	556	0.1	0.0	409623376	3888	??	Ss	8:28AM	1:24.16	/usr/local/mysql
root	8221	0.0	0.1	408231360	6832	??	Ss	10:32PM	0:00.11	/usr/libexec/sym
root	8220	0.0	0.1	408220176	5280	??	Ss	10:32PM	0:00.19	/usr/bin/sysdiag
saiabburi	8218	0.0	0.1	408243680	12208	??	S	10:32PM	0:00.30	/System/Library/
saiabburi	8217	0.0	0.1	408197200	5616	??	S	10:32PM	0:00.07	/usr/libexec/pro
_accessoryupdater	8210	0.0	0.1	408226624	5552	??	Ss	10:32PM	0:00.05	/System/Library/
saiabburi	8207	0.0	0.1	408207008	6688	??	S	10:32PM	0:00.15	/usr/libexec/bio
root	8206	0.0	0.1	408228624	12336	??	Ss	10:31PM	0:00.52	/usr/libexec/Pow
_accessoryupdater	8202	0.0	0.1	408226624	5200	??	Ss	10:31PM	0:00.09	/System/Library/
saiabburi	8200	0.0	0.1	408243856	10176	??	S	10:31PM	0:00.36	/System/Library/
saiabburi	8201	0.0	0.2	408229232	15328	??	S	10:31PM	0:00.18	/System/Library/
saiabburi	8198	0.0	0.1	408685760	5072	??	Ss	10:31PM	0:00.09	/System/Library/

Top :

(base) saiabburi@Geyasree-MacBook-Air OSSP % top

Processes: 392 total, 2 running, 390 sleeping, 2110 threads																												
Load Avg: 3.08, 2.16, 2.14 CPU usage: 1.83% user, 2.52% sys, 95.63% idle SharedLibs: 318M resident, 70M data, 14M linkedin. MemRegions: 308122 total, 2233M resident, 108M private, 976M shared.																												
PhysMem: 7538M used (1426M wired, 2599M compressor), 79M unused. VM: 168T vsize, 4283M framework vsize, 124450(0) swapins, 250511(0) swapouts. Networks: packets: 20513660/6801M in, 1462995/498M out.																												
Disks: 4902202/546 read, 4895819/346 written.																												
PID	COMMAND	%CPU	TIME	#TH	#WQ	#PORTS	MEM	PURG	CMPRS	PGRP	PPID	STATE	BOOSTS	%CPU_ME	%CPU_OTHS	UID	FAULTS	COW	MSGSENT	MSGRECV	SYSBSD	SYSMACH	CSW	PAGEIN	IDLEW	POWE	INSTRS	
8330	top	7.5	00:08.56	1/1	0	29	8753K	0B	2656K	8330	8285	running	*0[1]	0.00000	0.00000	0	41045+	72	1717214+	858001+	190043+	1198639+	4478+	29	2+	7.5	205208350	
372	WindowServer	6.5	78:13.72	18	4	2091-	645M-	8032K-	129M	372	1	sleeping	*0[1]	0.05370	0.16669	88	7472364+	22308	31537656+	31208545+	51264535+	103088974+	23816549+	4088	138700+	6.6	70483861	
0	kernel_task	6.1	68:58.52	464/8	0	0	15M	0B	0B	0	0	running	0[0]	0.00000	0.00000	0	37920	0	46568692+	37274074+	0	0	293174598+	0	10010449+	0.0	36048928	
6568	Google Chrom	2.6	06:15.43	37	1	459	739M-	0B	407M-	984	984	sleeping	0[55]	0.00000	0.00000	501	2625142+	61222	1538741+	906953+	2459257+	3059609+	2110293+	592	15823+	2.8	14456179	
8024	com.apple.We	1.6	00:20.30	6	2	88+	156M+	0B	40M-	8824	1	sleeping	*0[12266+]	0.00000	0.00000	501	76594+	180	1688304+	911169+	1047410+	4238925+	321461+	150	247	1.6	49482471	
1445	Google Chrom	1.2	06:41.83	31	1	394	353M+	0B	256M-	984	984	sleeping	0[43]	0.00000	0.00000	501	2314961+	30008	1562633+	492453+	1197484+	3163399+	2013219+	2831	6829	1.2	85964085	
8021	app.inkwell	1.2	00:15.92	18	2	334+	29M+	0B	13M	8021	7986	sleeping	*0[89+]	0.00000	0.00000	501	53022+	253	258288+	108611+	590671+	1239458+	521703+	197	153	1.2	21307265	
8027	com.apple.We	0.8	00:18.62	8	4	101+	21M+	48K	11M-	8827	1	sleeping	*0[31]	0.00000	0.00000	501	7654+	146	1690541+	2909466+	3158913+	7021466+	636408+	69	211	0.8	32908534	
7873	Terminal	0.8	00:11.67	7	2	253	81M-	50M	15M	7873	1	sleeping	*0[149]	0.00572	0.00000	501	258732	321	94880+	24054+	132209+	263489+	79971+	817	35	0.8	58602512	
8032	com.apple.We	0.7	00:11.13	4	1	80+	155M-	0B	96M-	8832	1	sleeping	*0[6368+]	0.00000	0.00000	501	70172+	172	1127799+	596080+	674359+	2758002+	215716+	4	171	0.7	16984258	
7621	Google Chrom	0.6	02:32.64	21	1	234	403M	0B	363M-	984	984	sleeping	0[21]	0.00000	0.00000	501	613416+	13250	330891	208220	511209	573801+	350633+	1455	795	0.6	4606388	
8029	com.apple.We	0.5	00:01.65	5	2	81+	64M+	0B	21M-	8829	1	sleeping	*0[2812+]	0.00000	0.00000	501	28643+	173	21338+	12105+	27654+	69667+	24536+	8	29	0.5	9716595	
8031	com.apple.We	0.5	00:04.33	7	2	90+	124M+	0B	39M-	8831	1	sleeping	*0[6112+]	0.00000	0.00000	501	48840+	182	68807+	39483+	67558+	216256+	46961+	174	68	0.5	11080762	
8030	com.apple.We	0.5	00:01.95	5	2	81+	117M-	0B	62M-	8830	1	sleeping	*0[2943+]	0.00000	0.00000	501	33278+	173	31027+	17685+	34458+	91224+	26278+	20	37	0.5	7599087	
904	Google Chrom	0.4	26:57.94	45	1	1290-	355M-	0B	192M-	984	1	sleeping	*0[7125]	0.00000	0.00000	501	7455632+	14740	28425197+	7672614+	42158467+	78628010+	17383252+	80653	10273	0.4	2517155	
8337	screencaptur	0.4	00:01.11	2	1	66	7666K	752K	0B	622	622	sleeping	*0[995+]	0.01365	0.00000	501	15278+	111	14072+	4725+	4547+	26598+	6497+	101	0	0.4	3835322	
7921	Google Chrom	0.4	00:01.59	18	1	131	58M-	0B	42M-	984	984	sleeping	0[10]	0.00000	0.00000	501	23448+	876	11150+	9591+	20315+	25734+	15008+	24	6	0.4	2093209	
556	mysqld	0.2	01:24.68	38	0	63	390M	0B	388M	556	1	sleeping	*0[1]	0.00000	0.00000	74	37038	106	115670+	57811+	8323419+	322336+	4120353+	1608	195267+	0.6	1821526	
366	corebrightne	0.1	01:46.83	3	2	129	3393K	0B	1392K	366	1	sleeping	*0[1]	0.16097	0.00000	0	169855+	107	422351+	286674+	2294006+	948308+	992417+	474	2241	0.1	1335921	
925	Google Chrom	0.1	17:19.93	17	1	167	85M	0B	43M	984	984	sleeping	*0[3]	0.00000	0.00000	501	951176	1932	2309480+	1296458+	128156205+	4116723+	16203862+	5184	9125	0.1	505437	
7986	Grammarly De	0.1	00:14.10	12	2	292	48M	144K	20M	7986	1	sleeping	*0[99+]	0.00000	0.00000	501	29650	668	170164+	26452+	235803+	333461+	208128+	1673	188+	0.2	828401	
7645	fileprovider	0.1	00:03.09	6	5	120	12M	0B	9696K	7645	1	sleeping	*0[357]	0.00000	0.00000	501	9772	218	3564	1290	75826+	7543	39919+	1242	160	0.1	688238	
315	powerd	0.1	00:19.99	4	3	133	4193K	0B	1456K	315	1	sleeping	*0[1]	0.00000	0.05370	0	112805+	117	108077+	108393+	555391+	322229+	238496+	245	172	0.1	706529	
557	com.muse.aut	0.0	00:19.01	51	2	26967+	5681K	0B	3952K	557	1	sleeping	*0[1]	0.00000	0.00000	0	11154	88	268032+	121032+	728446+	309157+	468345+	498	938	0.0	424773	
486	audioclocksy	0.0	00:15.70	2	1	63	4769K	0B	3072K	486	1	sleeping	*0[1]	0.00000	0.00000	0	18033	78	42153	245327+	228986+	478770+	240652+	93	14	0.0	253660	
8338	screencaptur	0.0	00:00.17	4	2	147	7346K	0B	80	8338	1	sleeping	*0[836+]	0.00000	0.01365	501	2002	170	1516+	989+	5506+	5643+	2266+	116	0	0.0	254593	
6756	Google Chrom	0.0	16:53.51	21	1	287	732M	0B	662M	984	984	sleeping	0[17]	0.00000	0.00000	501	2920532	23233	2335099	1002533	7559322	5174087+	6697187+	754	10842	0.0	61796	
2131	PerfPowerSer	0.0	01:50.30	5	3	323	13M	256K	3104K	2131	1	sleeping	0[2179]	0.00000	0.00000	0	135408	160	1090089	916490+	1290789+	1574377	988368+	1470	179	0.0	162094	
807	Google Drive	0.0	00:47.18	120	1	512	61M	0B	40M	700	700	sleeping	*0[1140]	0.00000	0.00000	501	373547	822	111688	34620	1446263+	413756	492152+	10549	9778	0.0	71462	
7985	mobiletimere	0.0	00:00.00	3	2	56+	2545K+	0B	1536K	7985	1	sleeping	*0[1]	0.00000	0.00000	501	1690+	90	228+	57	1875+	613+	569+	101	2	0.0	133359	
3475	AMPLibraryAg	0.0	00:03.17	5	1	130+	9089K+	0B	6320K-	3475	1	sleeping	*0[289]	0.00000	0.00000	501	283960+	223	4983	1924	38376+	10897	22332+	1305	154	0.0	147479	
685	sharingd	0.0	00:33.50	4	1	321	15M	0B	13M	685	1	sleeping	*0[1]	0.00000	0.00000	501	102335	270	208783+	62267	658965+	756828+	328374+	2731	182	0.0	1069649	
320	logd	0.0	01:42.01	4	3	1272	26M	0B	26M	302	1	sleeping	*0[1]	0.00000	0.00000	0	276809	82	659360+	175183+	3512927+	1572665+	28193	256	0.0	232963		
309	mds	0.0	00:50.74	74	5	328+	38M-	0B	34M-	329	1	sleeping	*0[1]	0.00000	0.00000	0	226702+	133	58055	46107	1769395+	161107+	780688+	944	10033	0.0	118372	